

TRACEY (traceyDB) — Architecture & Maintenance Manual

This document explain the project's architecture, the main views, the admin panel (features.html), and the frontend structure, so that anyone picking up the project can maintain it without having to reread the whole codebase from scratch.

Running the dev server

```
/path/to/env/with/django/bin/python manage.py runserver localhost:8000
```

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1. General architecture

1.1 Repository layout

core/	Django project (settings.py, urls.py, wsgi.py, asgi.py)
apps/	only app registered in INSTALLED_APPS ("apps.config.AppsConfig")
— home/	the "real" app: views, forms, models, plots, utils
— authentication/	login/register/logout
— management/commands/	custom manage.py commands (see section 5)
— templates/	layouts, includes, menus, home/ pages (see section 4)
— static/	CSS/SCSS, JS, images (see section 4)
utils/	bioinformatics scripts (HMM, BLAST, NCBI taxonomy, phylogenetic trees)
	NOT a Django app, these are scripts invoked by the management commands
docs/	this manual

Prior diagrams (not duplicated here, useful as visual reference): traceyStructure.pdf/pptx, tracey_all_models.pdf, tracey_main_models.pdf, tracey_other_models.pdf, DomaingroupsStructure.pdf/pptx, tree.pdf (all at the repo root).

1.2 The INSTALLED_APPS quirk

In core/settings.py, INSTALLED_APPS only registers 'apps.config.AppsConfig' — it does **not** register apps.home or apps.authentication as separate apps. Since Django's app registry assigns models by module-path prefix matching, models defined in apps/home/models.py end up under the apps app-label. That's why apps/admin.py registers the admin by iterating apps.get_app_config('apps') instead of apps.get_app_config('home'). The HomeConfig class in apps/home/apps.py exists but, since it's not in INSTALLED_APPS, is effectively dead code.

1.3 Database

The backend is MySQL (core/settings.py, ENGINE: django.db.backends.mysql), pointing at a **pre-existing** schema. All models in apps/home/models.py have managed = False. This means:

- Schema changes are **not** done with makemigrations/migrate. The MySQL database has to be modified directly (or via the .sql dumps in utils/), and the change then reflected by hand in models.py.
- apps/migrations/ only has 0001_initial.py — there is no real migration history to maintain.

Backups: the server keeps up to 30 daily backups of the database (one per day; once the cap is reached, the oldest backup is deleted as each new one is created). To work on a local copy, download the most recent backup from the server and restore it into a local MySQL instance:

1. Copy the latest backup file from the server (SCP/SSH):

```
scp <user>@<server>:/path/to/backups/<latest-backup>.sql ./utils/
```

2. Create a local database and import the dump:

```
mysql -u <local_user> -p -e "CREATE DATABASE tracey_local CHARACTER SET utf8mb4"
mysql -u <local_user> -p tracey_local < utils/<latest-backup>.sql
```

3. Point .env (repo root) at the local database — DB_NAME, DB_USER, DB_PASSWORD, DB_HOST=localhost, DB_PORT (see core/settings.py, DATABASES).

4. Run the dev server as usual (see the top of this document) — Django will connect to the local copy instead of the server's database.

Downloaded dump files should **not** be committed — .gitignore already excludes utils/*.backup.sql and utils/tracey_20180606_0400.dump2.

1.4 Routing (core/urls.py → apps/home/urls.py)

core/urls.py includes, in order: admin/, apps.authentication.urls (login/register/logout), captcha/, robots.txt, sitemap.xml, and **last** apps.home.urls. A comment in the file itself warns that apps.home.urls must stay last, because it ends with a catch-all:

```
re_path(r'^.*$', views.pages, name='pages')
```

The pages() view (in apps/home/views_query.py) takes the last URL segment and loads home/<segment>.html directly as a template — that's how standalone pages like /contact.html work without an explicit route. updates.html is gated behind login via the _PAGES_REQUIRES_LOGIN set.

1.5 The “grouped views” pattern

Views are split across several modules inside apps/home/, grouped by functional family, and views.py acts as a **single aggregator**: it does explicit named imports from each module and re-exports them. urls.py only does from apps.home import views and calls views.<Name> — it never imports a sub-module directly.

Module	View family
apps/home/views_query.py	query-sequences, query-sequences-results, FASTA/multialignment export, 3D viewer, query-sequence-details, ajax endpoints for autocomplete and cascading dropdowns
apps/home/views_motifs.py	query-motifs, query-motifs-results, and the motifScan helper (shared HMM-scanning engine)
apps/home/views_verify.py	query-insert, query-verify (curation menu and view), query-verify-blast
apps/home/views_trees.py	Phylogenetic tree view (trees)
apps/home/views_admin.py	features.html and all its actions (see section 3)

If a new view is added, it must be added both in its corresponding views_*.py module and in the explicit import list in apps/home/views.py so it's exposed to urls.py.

1.6 The segment convention (menu highlighting)

Each view manually computes a segment context variable (typically request.path.split('/')[-1] or a variant) and passes it to the template. includes/sidebar.html uses that variable in conditionals such as {% if 'query' in segment %} to highlight the active menu item.

There is no central context processor for this — it's a pattern copy-pasted view by view. If a new view is added and segment is forgotten, the sidebar highlighting simply won't work for that page (a silent failure, no error).

1.7 Key models

All in apps/home/models.py, managed = False (see the model PDFs at the repo root for the full schema):

Model	Purpose
Sequences	Central entity: a protein/gene sequence (annotation, status, taxonomy, associated gene)
Motifs	Verified/active HMM motif hits on a sequence
Verifymotifs	Mirror of Motifs for hits pending verification (the query-verify workflow)
Domaingroups	Functional/structural domain groups, with a parent hierarchy (domaingroupparent_id, can have several parents separated by ;)
Domains	Top-level protein domain (SNARE, Habc, Longin, LGL, ...)
Taxonomies	Node in TRACEY's own taxonomy tree (scientific name, rank, parent)
Methods	How a motif/verifymotif was generated (type = 'hmm' or 'manual')

2. Reference of the main views (query-*)

QuerySequences / QuerySequencesResults

File: apps/home/views_query.py. URLs: query-sequences, query-sequences-results.

- QuerySequences renders the search form (home/query-sequences.html). Key context: domainsList, proteinLayoutsList (from get_menu(request), distinguishes staff vs public menu), shortnames (cached 24h), taxonomy_ranks, form (FamilyForm), is_staff, error (session message if the previous search returned 0 results).
- QuerySequencesResults runs the actual search by delegating to get_sequences() (apps/home/utls.py), which combines non-motif filters (aliases, status, species, taxonomy subtree via BFS) with motif filters (domain/domaingroup/proteinlayout), and splits results into verified (Motifs) vs unverified (VerifyMotifs). Hard cap of 4000 rows. Key context: sequences, speciesname, motifnames, hmmModels (list of available .hmm files for generating an MSA).

QuerySequencesFastaFormat

URL: query-sequences-fasta (POST).

Export endpoint with 4 modes depending on which POST field is present: TSV table download, plain per-sequence FASTA, per-motif FASTA (extracted from the XML in asciiooutput), or HMM multialignment (via pyhmmer.hmmalign) to view/download.

QuerySequences3dViewer

URL: query-sequences-3dViewer/<sequence_id>.

Renders the 3D viewer (3Dmol.js) for a sequence. Key context: sequence, fetch3d (AlphaFold URL), motif_coords (real coordinates of each domaingroup, recomputed against the full sequence). Uses user_can_access_sequence() (apps/home/utls.py) to verify that all of the sequence's domaingroups are within the menu allowed for the caller (staff vs public); raises Http404 otherwise.

QuerySequencesDetails (query-sequence-details)

URL: query-sequences/<sequence_id>/details/.

Sequence detail page. Key context: sequence, speciesname, wiki_image, pdb/pdb_name (parsed out of foreignannotation), fetch3d, layout (SVG plot via build_domain_plot()), and motifs — a dict per active Motifs entry with its domaingroup-parent chain, fields parsed out of the asciiooutput XML (consensus, similarity, e-value), and an individual plot per motif. Same access check as the 3D viewer.

DetailsSequencesFastaFormat

URL: details/fastaFormat/<sequence_id>/. Simple FASTA download of a sequence, with the same access control.

QueryMotifsView / QueryMotifsResultsView

File: apps/home/views_motifs.py. URLs: query-motifs, query-motifs-results.

Form to scan a hand-pasted sequence against TRACEY's HMMs (no need for the sequence to already exist in the database). QueryMotifsView validates the input (non-empty, ≤2000 aa) and redirects to QueryMotifsResultsView, which calls the motifScan() helper.

motifScan() (same file) is the HMM-scanning engine also reused by QueryInsertView and QueryVerifyView: it validates the input isn't FASTA-formatted, runs pyhmmer against the relevant subset of .hmm files (or the full MOTIFS.hmmDb if proteinlayout == "ALL"), filters by evalcutoff, resolves domaingroup/domain names and real alignment coordinates, and returns hits_d (a hits dict sorted by ascending p-value).

QueryInsertView (query-insert)

File: apps/home/views_verify.py. Staff-only (@login_required + @staff_login_required).

Form to insert a new sequence (InsertSequence ModelForm). On save, it creates the Sequences row, immediately runs motifScan(..., ["ALL"]), persists the hits as VerifyMotifs via saveVerifyMotifs(), and redirects to query-verify for that sequence. If saving or scanning fails, it rolls back by deleting the newly created Genes row.

QueryVerifyMenuView / QueryVerifyView (query-verify)

File: apps/home/views_verify.py. Staff-only.

- QueryVerifyMenuView (query-verify, no id) is the listing/menu page for sequences pending curation; the actual table is loaded via ajax through load_queryverifysequences (apps/home/views_query.py, partial template home/query-verify-update-sequences.html).
- QueryVerifyView (query-verify/<sequence_id>) is the curation workbench view, the most complex view in the project. From the same form you can: verify/promote a VerifyMotifs to Motifs (or the reverse, "unverify"), delete motifs or the whole sequence, re-scan (motifScan + saveVerifyMotifs), or add a manual motif (add_manual_motif, creates a Methods row with type='manual'). Each action appends an audit line to the sequence's changelog field.

QueryVerifyBlastView (query-verify-blast)

URL: `query-verify/traceyBLAST/<db>/<query_id>`.

Compares a sequence/motif/verifymotif against one of TRACEY's local BLAST databases (or redirects to public NCBI BLAST if db indicates so), using the local `blastp` binary at `utils/ncbi-blast-2.13.0+/bin/blastp`, and renders the result as a colored alignment viewer.

Supporting ajax endpoints

A set of small GET views in `views_query.py` (`load_taxonomy_rank`, `load_species`, `load_domains`, `load_domaingroups_rank1/2`, `load_sequenceshortnames`, `load_taxonomy_by_shortname`, `suggest_aliases`, `updateSequenceStatus`) and in `views_verify.py` (`suggestNames`, `autocompleteModel`) feed the cascading dropdowns and autocomplete of the search/insert/verify forms. They're not detailed one by one here — they all follow the same pattern: receive a parameter from the level above and return partial HTML or JSON with the options for the next level.

3. features.html — admin panel

Template: `apps/templates/home/features.html`. View: `features()` in `apps/home/views_admin.py`. Staff-only.

This is TRACEY's data-maintenance panel: six cards, each with its own form + ajax call + inline script at the bottom of the file.

Feature	What it does	Ajax endpoint	Script/command behind it
Update Taxonomies	Refreshes the taxonomy table against current NCBI data; creates new taxa if needed. Takes ~10 min.	<code>ajax_update_taxonomy</code> → <code>views.update_taxonomy</code>	<code>utils/ncbi_taxonomy/TaxonomyUpdater.py</code> , same engine as <code>manage.py UpdateTraceyTaxonomies</code>
Update TRACEY Sequences	Re-syncs "live" sequences that have an NCBI ID against current data (sequences without an NCBI ID are left untouched). Filterable by species/family. Can take hours.	<code>ajax_update_sequences</code> → <code>views.update_sequences</code>	<code>utils/traceySequenceUpdater/traceySequencesUpdater.py</code> , same engine as <code>manage.py UpdateTraceySequences</code>
Re-scan Motifs	Re-runs the HMM scan over existing sequences to refresh Motifs/Verifymotifs. Filters: species, family/domain/specific HMM, e-value threshold, "only active", dry-run mode (opens the result in a new tab without persisting).	<code>ajax_rescan_motifs</code> → <code>views.rescan_motifs</code>	<code>utils/motifPredictor/reScanMotifs.py</code> , same engine as <code>manage.py ReScanMotifs</code>
Upload new sequences	Uploads a FASTA of new sequences; TRACEY analyzes them against the HMMs and auto-assigns the best-hit motif.	<code>ajax_upload_sequences</code> → <code>views.upload_sequences</code>	Required FASTA header format documented in <code>home/help_doc_upload_sequences.html</code>
HMM Models (download)	Downloads TRACEY's HMM profiles (one family or the full database) for external use with <code>hmmsearch/hmmscan</code> .	<code>download_hmm_zip</code> (direct navigation, not ajax)	Catalogs <code>.hmm</code> files in <code>utils/hmmModels/</code> via <code>get_hmm_catalog()</code>
Build Phylogenetic Tree	Rebuilds the tree of life for all taxonomies currently in TRACEY, following current NCBI phylogeny. Provides download links for the tree in Newick and PDF form.	<code>ajax_update_tree</code> → <code>views.update_tree</code>	<code>utils/ncbi_taxonomy/TreeUpdater.py</code> , same engine as <code>manage.py UpdateTraceyTree</code>

Common pattern across all six: a CSRF token in each form, a spinner + status message, and a shared flag (`reloadOnAjaxStop`) that reloads the page once any of these long ajax calls finishes, to refresh the "last updated"

timestamps.

4. Frontend: templates, JS, includes and menus

4.1 Template tree (apps/templates/)

layouts/	
base.html	main layout (sidebar + content + footer)
sequenceForm.html	large insert/verify-sequence form (an include, not a base)
includes/	
sidebar.html	navigation menu (hardcoded, see 4.3)
footer.html	page footer (partner logos)
scripts.html	common JS bundle (Popper, Bootstrap, Chartist, SweetAlert2, moment.js, volt.js, ...)
settings-box.html	Volt theme widget (commented out in base.html, unused)
accounts/	login.html, register.html
home/	~30 page templates + partials/_motif_card.html
menus/	NOT HTML templates – Python modules (see 4.3)

layouts/base.html defines the blocks title, og_title, og_description, stylesheets, content, javascripts. Most templates in home/ and accounts/ extend it directly (a single level of inheritance, no deeper chains).

Some templates in home/ are **not full pages**: they’re HTML fragments returned directly by ajax views (e.g. query-sequences-fasta.html, query-verify-update-sequences.html, dropdown option lists). Others are standalone HTML documents opened in a new tab (query-sequences-3dViewer.html, query-verify-blast.html, each with its own <!DOCTYPE html>).

home/partial/_motif_card.html is reused twice in layouts/sequenceForm.html: once for verifymotifs (with a Verify/Delete option) and once for motifs (with an Unverify/Delete option).

4.2 Main JS files (apps/static/assets/js/)

File	Role
volt.js	JS for the Volt theme (Bootstrap 5): SweetAlert2, settings panel, general theme chrome
sequenceForm-dropdowns.js	Cascading dropdowns proteinlayout → domain → domaingroup → domainsubgroup on the insert/verify form (and its “manual” variant)
sequenceForm-utils.js	suggestNames() (ajax to ajax_suggestNames) and searchOpen() (taxonomy autocomplete via /search.json)
ngl-viewer.js	3D viewer based on NGL.js , used on the insert/verify sequence form (colors by domain, highlights residues)
django-htmx.js	Debug helper for django_htmx: when DEBUG=True, replaces HTMX error responses with Django’s debug error page
jquery-4.0.0.min.js	jQuery, loaded per-page (not globally from base.html) wherever AJAX/DOM manipulation is needed

Important: the standalone detail/3D-viewer pages (query-sequences-3dViewer.html, query-sequences-details.html) instead use **3Dmol.js via CDN** (<https://3Dmol.org/build/3Dmol-min.js>). In other words, TRACEY has **two different 3D viewers** depending on the page: NGL.js (local) in the insert/verify workflow, and 3Dmol.js (CDN) on the public query pages. If you touch 3D-visualization functionality, check both.

4.3 How TRACEY builds its menus

There are two distinct concepts both loosely called “menu” — worth not confusing:

1. **Navigation menu (sidebar)** — includes/sidebar.html. This is **hardcoded** HTML, not generated from the database or from a Python data structure. Active-item highlighting is done with Django conditionals ({% if 'query' in segment %}) over the segment variable each view computes by hand (see 1.6). The only real context processor in play is apps/context_processors.py:cfg_assets_root, which just injects ASSETS_ROOT for static-asset paths.
2. **“Menu” of protein families** — apps/templates/menus/query_sequences.py and query_sequences_full.py. Despite living under templates/ and having a .py extension, **these are not Django templates**: they’re Python modules holding a nested dict (menu = {...}) describing the protein-family → domain → domaingroup hierarchy (SNARE, Ras, AAA, ...), plus helper functions (get_keys_recursively(), get_dict()). Views import them to populate the cascading dropdowns on Query/Insert/Verify/Features. Unrelated to UI navigation.

4.4 CSS / SCSS

Base theme “Volt” (Bootstrap 5, AppSeed/Themesberg), compiled from SCSS (apps/static/assets/scss/) to CSS (apps/static/assets/css/volt.css). Project-specific customization in scss/custom/_variables.scss and css/sequenceForm.css (styles for the insert/verify form). The CSS specific to features.html (loaders, .feature-card grid) is inline in the template itself rather than in a separate file.

5. Maintenance commands (manage.py)

Defined in `apps/management/commands/`. Each one has an equivalent feature in `features.html` (section 3) that triggers the same logic from the UI:

Command	What it does
<code>UpdateTraceySequences</code>	Refreshes “live” sequences that have an NCBI ID. Flags: <code>--continue</code> , <code>--force</code> , <code>--onlyActive</code> , <code>--species</code> , <code>--traceyIds</code> , <code>--domain</code>
<code>UpdateTraceyTaxonomies</code>	Refreshes taxonomies from NCBI. Flag <code>--taxa</code> (default <code>superkingdom</code>)
<code>UpdateTraceyTree</code>	Rebuilds the phylogenetic tree
<code>ReScanMotifs</code>	Re-scans HMMs over existing sequences. Requires <code>--hmm</code> or <code>--family</code>
<code>plotTaxonomy</code>	Diagnostic command: draws the taxonomy tree/network for a given scientific name (not part of the data pipeline)

Additionally, `utils/traceySequenceUpdater/updateDomainGroupsWithHMMs.py` is a standalone maintenance script — not a `manage.py` command — run via:

```
python manage.py shell < utils/traceySequenceUpdater/updateDomainGroupsWithHMMs.py
```

It syncs new HMM files under `utils/hmmModels/` into the rest of the system: adds any missing entries to the `query_sequences_full.py` menu, creates the corresponding `Domaingroups` rows in the database, and rebuilds `utils/hmmModels/MOTIFS.hmmDb` (concatenates every `.hmm` file and re-indexes with `hmmpress -f`, via `rebuildMotifsHmmDb.py`) SO `motifScan(proteinlayout="ALL")` picks up the new profiles. It does **not** touch the public `query_sequences.py` menu — that still has to be edited by hand (see 7).

6. Access control (staff permissions)

The database-management sections — `features.html`, `query-insert`, `query-verify` (and its BLAST sub-view) — are gated behind `@login_required + @staff_login_required` (`apps/home/views_verify.py`, `apps/home/views_admin.py`), which check `request.user.is_staff`. A plain registered user has **no** access to these sections; only users flagged as staff do.

To grant access to a user who has already registered:

1. Go to `tracey.unil.ch/admin` (Django admin) and log in with a superuser/staff account.
2. Open the **Users** section and select the target user.
3. Check “Staff status” (and any relevant permissions/groups) and save.

Careful not to confuse two different models both called “User”: the one you edit here is Django’s standard `auth.User` model (the one `/admin` manages). There’s also a legacy `User` table in `apps/home/models.py` (`managed = False`), which is a pre-Django authentication table unrelated to the current access control — that’s not the one to touch to grant permissions.

7. Adding a new HMM model and generating its Domaingroups

Adding a new HMM profile touches three things that all have to stay in sync: the `.hmm` file itself, the `Domaingroups` row in the database, and the domain menu dicts that drive the cascading dropdowns on `Query/Insert/Verify/Features`. Most of this is automated by `utils/traceySequenceUpdater/updateDomainGroupsWithHMMs.py` (see 5) — this section explains what it does step by step so a new HMM can be added correctly.

7.1 Drop the `.hmm` file in the right place

Add the file to `utils/hmmModels/<FOLDER>/`, where `<FOLDER>` is the family folder already mapped in `DOMAIN_CONFIG` at the top of `updateDomainGroupsWithHMMs.py` — e.g. `SNARE`, `HABC`, `LONGIN`, `LGL`, `C2`, `AAA.AAA`, `AAA.ND`, `RAS`, `ARF`, `MUN.D1/MUN.D2`, `NSR.CD/NSR.MD/NSR.ND`, `PROPPIN`, `RHOMBOLD`, `RINT`, `SM.D1/SM.D2A/SM.D2B/SM.D3`, `SNAP`, `ZW10`. If the HMM belongs to a brand-new family with no existing folder, create the folder and add a new entry to `DOMAIN_CONFIG` (`folder - (domain_name_in_DB, menu_path_list)`) before running the script — otherwise it will be skipped.

The file’s basename (without `.hmm`) becomes both the `Domaingroups.domaingroupname` and the menu key, so naming matters:

- If the name matches (case-insensitive) a menu key that already exists, or a per-folder alias defined in `DOMAIN_CONFIG` (e.g. `SNAP` maps `aSnap/cSnap` → `aSNAP/cSNAP`), it’s treated as already present and nothing new is added.
- If the name is listed in `HMM_BLACKLIST`, it’s skipped entirely — use this for general HMMs that shouldn’t get their own menu entry (e.g. the `SM Vps33/Vps45` base HMMs, superseded by the `Vps33a/Vps33b` variants already in the menu).
- Dot notation infers hierarchy: `Longin.V` is nested as a child of `Longin` in the menu if a `Longin` key already exists there; otherwise it’s added at the top of the family subtree.

7.2 Run the sync script

```
python manage.py shell < utils/traceySequenceUpdater/updateDomainGroupsWithHMMs.py
```

This does everything else automatically:

1. **Menu sync** — `sync_menu_with_hmms()` scans `utils/hmmModels/` and adds any `.hmm` not already represented (per the rules in 7.1) to the **staff** menu, `apps/templates/menus/query_sequences_full.py` (see 4.3).
2. **Database** — `updateDomainGroups()` walks the (updated) menu and, for every key without a matching `Domaingroups` row, creates one linked to its parent `Domains/Domaingroups`, reading `domaingrouplength` from the `LENG` field of the corresponding `.hmm` file. Motif scanning (`motifScan()`, `apps/home/views_motifs.py`) resolves each HMM hit against a `Domaingroups` record, so a `.hmm` with no matching row can't be classified/displayed correctly.
3. **MOTIFS.hmmDb rebuild** — `rebuild_motifs_hmmdb()` (`rebuildMotifsHmmDb.py`) concatenates every `.hmm` under `utils/hmmModels/` into `utils/hmmModels/MOTIFS.hmmDb` and re-indexes it with `hmmcompress -f`, so the new profile is picked up by `motifScan(proteinlayout="ALL")` scans, not just family-scoped ones.

7.3 What's still manual

`apps/templates/menus/query_sequences.py` — the **public** (non-staff) menu — is **not** touched by the script. If the new `domaingroup` should be visible to non-staff users, add it there by hand (see 4.3, and 6 for the staff/public distinction).

8. Notes and gotchas for the next developer

- **Database schema changes are NOT done with `makemigrations/migrate`** — all models are managed = `False` over a pre-existing MySQL database (see 1.3).
 - `HomeConfig` (`apps/home/apps.py`) is not in `INSTALLED_APPS` — appears to be dead code.
 - TRACEY has **two different 3D viewers** (`NGL.js` vs `3Dmol.js` via CDN) depending on the page — see 4.2.
 - Menu highlighting depends on every new view manually defining segment — there's no automatic mechanism for this (see 1.6).
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9. Note on this document and exporting to PDF

This manual is deliberately kept in Markdown: it's versioned alongside the code, editable without special tooling, and renders directly on GitHub/GitLab/VSCoDe. If a PDF version is ever needed, it can be generated with:

```
pandoc docs/MANUAL_en.md -o manual.pdf
```